

Mitigation / Preventive Actions for Risks

Introduction

After identifying potential risks in the StockWise inventory management system, it is important to define **preventive and mitigation actions**. These actions aim to reduce the likelihood of risks occurring or minimize their impact if they do occur.

Mitigation Strategies for Identified Risks

1. Data Loss Risk

- Implement regular automatic database backups (daily or real-time replication)
- Use transactional database operations to prevent partial updates
- Ensure secure storage and disaster recovery plans

2. Security Risk

- Apply role-based access control (RBAC) to limit user permissions
- Use strong password policies and BCrypt hashing for authentication
- Enable HTTPS/TLS encryption for all data transmissions
- Conduct regular security audits and vulnerability scanning

3. System Performance Risk

- Monitor server and application performance continuously
- Optimize database queries and API endpoints
- Implement caching mechanisms for frequent requests
- Conduct load testing to ensure system handles expected traffic

4. Integration Risk

- Clearly document API endpoints and data contracts
- Perform integration testing between frontend and backend systems
- Use version control and staging environments to test updates before production

5. User Error Risk

- Design a clear and user-friendly interface with guidance and tooltips
- Implement confirmation prompts for critical actions (e.g., deleting products)
- Provide user training and updated documentation
- Maintain audit logs to track changes made by users

6. Server or Deployment Risk

- Use reliable hosting services and redundant servers if needed
- Set up monitoring and alerting systems for downtime or failures
- Create automated deployment scripts to reduce human errors

- Prepare a rollback plan for deployment failures

Conclusion

By applying these mitigation and preventive actions, the StockWise project aims to:

- Reduce the probability of critical failures
- Minimize the impact of risks on system performance and data integrity
- Ensure a reliable and secure inventory management platform for users